

The thermal expansion of aluminium from 0° to 650° C

To cite this article: A J C Wilson 1941 *Proc. Phys. Soc.* **53** 235

View the [article online](#) for updates and enhancements.

You may also like

- [X-ray line broadening study of splat cooled aluminium and aluminium-tin and aluminium-nickel solid solutions](#)
A Kirin and A Bonefacic
- [A General-Purpose Debye-Scherrer Camera and its Application to Work at Low Temperatures](#)
William Hume-Rothery and D J Strawbridge
- [Lattice parameters of iron-aluminium alloys](#)
J A von Fraunhofer

THE THERMAL EXPANSION OF ALUMINIUM FROM 0° TO 650° c.

By A. J. C. WILSON,
Cavendish Laboratory, Cambridge

Received 5 December 1940. Communicated by Prof. Sir W. Lawrence Bragg, F.R.S.

ABSTRACT. The lattice spacing of aluminium has been measured as a function of temperature by means of a high-temperature Debye-Scherrer x-ray camera. The coefficient of expansion has been deduced. The results are :

Temp.	0°	100°	200°	300°	400°	500°	600°	650° c.
d	4.0391	4.0486	4.0592	4.0701	4.0820	4.0947	4.1087	4.1162
$10^6 \alpha$	22.0	25.4	26.5	27.8	29.9	32.5	35.5	37.2

The systematic and random errors in d are each about 0.0001 Å., the error in α from 1 to 2 per cent.

§ 1. INTRODUCTION

THE thermal expansion of aluminium has been investigated at the Reichsanstalt by Dittenberger (1902) and at the U.S. Bureau of Standards by Hidnert (1925). There are also numerous determinations covering shorter temperature ranges. The differences (5 to 10 per cent in the expansion coefficient) are sufficiently large to make further measurements of interest. This paper describes the results of work with a high-temperature Debye-Scherrer x-ray camera. Esser, Eilender and Bungardt (1938) have published x-ray results for several metals, including aluminium, but their accuracy (0.01 to 0.05 per cent in the lattice parameter) is considerably less than that attained in the present work.

§ 2. DESCRIPTION OF CAMERA

The camera used is a development of those of Jay (1933) and Hume-Rothery and Reynolds (1938). It consists of two circular brass plates, the upper (A in figure 1) about 19 cm. in diameter, the lower (B) somewhat larger and containing a groove to receive the cover C. A boss of the same diameter as the top plate is turned on the lower plate. The plates are held about 1.1 cm. apart by two brass V-blocks D and E. Each V-block carries a pair of stainless steel knife-edges which mark out definitely the length of film exposed to the x rays. The film, backed by a sheet of black paper, is held firmly against the upper plate and the boss of the lower plate by two strips of steel ribbon on each side. The strips are attached to the back V-block D and are clipped firmly to the front one E by

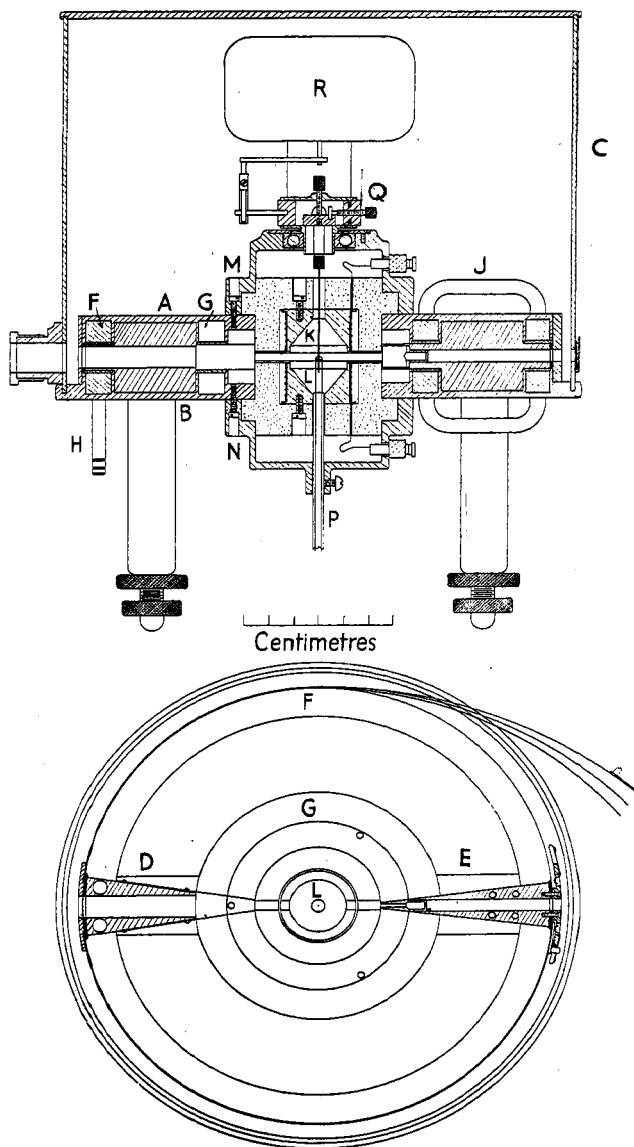


Figure 1. The camera.

phosphor-bronze springs. The lower diagram of figure 1 shows, on the left side of the camera, the film, paper and steel strips lying loosely; on the right side it shows them clipped to the front V-block.

Each plate contains two water-cooling channels F, G of about 1 cm. square section. The water enters through one of four tubes H at the back of the camera, passes through the V-block D to the upper plate, where it flows half-way round the outer channel till it meets a baffle, through the U-tube J to the inner channel, where it flows completely round and through another U-tube to the other half of the outer channel and finally emerges through another of the tubes H. The circulation in the lower plate is similar.

The furnace consists of two bobbins K, L of oxidation-resistant steel. The windings are of platinum insulated from the bobbins by mica. The bobbins are fastened by screws in two soapstone members held to the plates of the camera by circular brass clamps M, N. The leads from the furnaces are taken to small bakelite terminal boards on the clamps. The lower clamp N supports a platinum *vs.* platinum-rhodium thermocouple P which serves to measure the temperature of the furnace. The thermocouple is of the ring type described by Hume-Rothery and Reynolds (1938). The upper clamp M contains a ball-race carrying the device Q for centering the specimen. The specimen is contained in a thin-walled silica tube which is mounted in a brass rod screwed into a small brass block. This contains two mutually perpendicular grooves into which fit small discs borne on the ends of two screws. The specimen is easily brought into the axis of rotation of the camera by adjusting these screws. A third screw clamps the block in position after centering. The specimen is turned four times a minute by the small synchronous motor R, which can be swung back out of the way when it is necessary to remove the specimen holder. The leads from the motor and the upper half of the furnace pass out of the camera through four holes in the front V-block E.

The x rays enter the camera through a small hole in the front of the cover. Filters are put in a small pocket on the outside of the cover when necessary. The x rays pass through a hole in the block E and strike the slit system S, a small removable brass block fitting the inner end of the hole. The slit itself, about 0.8 by 3 mm., is at the outer end of this block; the wider part towards the centre of the camera prevents rays scattered by the edges of the slit from striking the film. The rays then pass out of the camera through a hole in the block D larger than that in the block E. Two strips fastened to the outside of D prevent any rays scattered by it from striking the film.

The heating currents in the two furnace bobbins can be varied independently, and at each temperature are adjusted so that the temperature distribution, when investigated by moving the thermocouple up and down, is symmetrical about the centre. Typical distributions are shown in figure 2. During the exposure the thermocouple is kept at the point marked T, and the reading corrected for the temperature difference between this point and the mean temperature of the effective part of the specimen. The temperature distribution is determined at the beginning of, twice during, and at the end of each exposure. At the higher

temperatures it is necessary to place an opaque screen between the furnace and the film to prevent fogging by the light emitted. The form of the temperature distribution depends somewhat on the material used for the screen. Black paper and aluminium foil have been used. Since aluminium foil reflects back much of the radiation from the furnace, the "centre drop" disappears when it is used. In every case the temperature distribution observed at the time is used in making the correction.

The furnace current is continually regulated by hand in order to keep the temperature constant. It rarely varies by more than 1° from the desired temperature, and the mean temperature during an exposure would differ from the

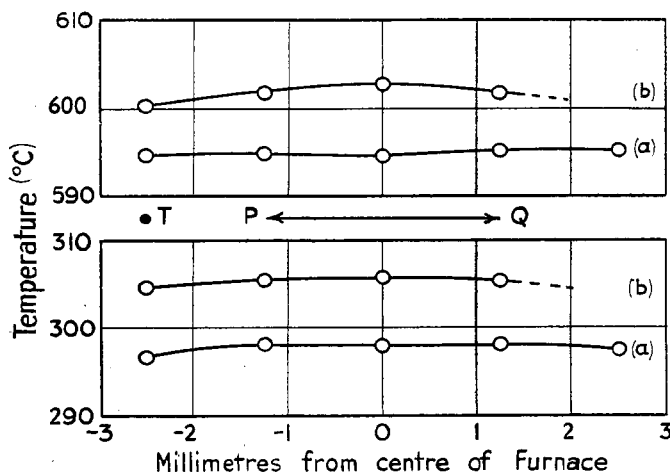


Figure 2. Temperature distribution in furnace.
T, position of thermocouple; PQ, effective part of specimen.
(a) black paper; (b) aluminium foil between furnaces and film.

desired temperature by less than this, perhaps about 0.5° . The camera has not yet been used for work above 660°C ., but was once maintained at 1000°C . for about half an hour without obvious damage.

§ 3. THE TEMPERATURE SCALE

Two platinum *vs.* platinum-rhodium thermocouples were used. Their e.m.f. was measured with a Cambridge Instrument Company Laboratory Thermocouple Potentiometer. Readings could be made to one microvolt without difficulty. The first thermocouple, protected by a silica tube, was calibrated by measuring its e.m.f. in melting and freezing tin, lead, zinc and sodium chloride. A curve was drawn of the deviation of its e.m.f. from the standard table given in the *International Critical Tables*. After a number of photographs of the aluminium specimen had been taken, short exposures were made near its melting point to determine the e.m.f. at which the reflections disappeared. This occurred about 6° higher than was expected. A similar test with lead showed its melting

point about 2° high. The temperatures of the exposures taken previously were recalculated using a deviation curve derived from the melting points of the lead and aluminium specimens. There was some possibility of the junction having been contaminated with zinc or aluminium, so a second thermocouple was made and calibrated in steam and at the melting points of lead, antimony and aluminium. The melting points of the metals were obtained by finding the e.m.f. at which their x-ray reflections disappeared. This method has the obvious advantage that conditions during calibration are the same as those during use. The e.m.f. could be determined within four microvolts. The melting points were taken to be

lead	327°·3 c.,
antimony	630°·5 c.,
aluminium	658°·9 c.

Nearly half the spacing measurements were repeated with the new thermocouple. The mean difference, after correction for unavoidable small differences in temperature, was -0.0001 A., which, even if entirely due to errors in the temperature measurement, corresponds to less than a degree difference between the two temperature scales. It is thought that the temperature measurements were reproducible within 0.5° and represent the International Temperature Scale within 1° .

Two objections have been raised against these temperature measurements. The first is that the temperature of the thermocouple is not that of the specimen. This may be true, but if so the difference is included in the calibration, since the temperature difference between the specimen and thermocouple is the same during calibration as during use. The second is that the temperature at which the x-ray reflections disappear might be higher than the melting point. This appears extremely unlikely. If the x-ray reflections from the liquid are the same as those of the solid, the liquid must have the same crystal structure and therefore the same physical properties as the solid.

§ 4. MEASUREMENT OF SPACINGS

The theory behind the use of this camera for the precise measurement of spacings has been considered fully by Bradley and Jay (1932). Some account of the practical details ought however to be given. As the camera is about twice the diameter of those described by Bradley and Jay, a piece of film encircling it completely would be of inconvenient size, so two pieces are used. To obtain S , the film length corresponding to four times the apparent angle of reflection, it is necessary to obtain the sum of three quantities:

- (1) the distance between the back knife-edge and the line on the left half of the film,
- (2) x , the arc length between the back knife-edges,
- (3) the distance between the back knife-edge and the line on the right half of the film.

Measurements are made with an instrument made by the Cambridge Instrument Company to a National Physical Laboratory design. It is described fully in a paper by Sears and Turner (1941). The film is placed in it so that the scale reading of the shadow of the back knife-edge is $x/2$. The scale reading when the cross-hair is set on a line is then $x/2$ plus the distance between the shadow of the back knife-edge and the line. S is the sum of the scale readings of the line on the two halves of the film. S_k is the sum of the scale readings of the shadows of the front knife-edges on the two halves, and the apparent angle of reflection is $\theta = S\theta_k/S_k$.

The determination of the angles of reflection involves the knowledge of two constants of the camera, x and θ_k . These were obtained from actual measurement of the relevant parts of the camera. It has been usual (Bradley and Jay, 1933) in work with cameras of this type to use a camera angle chosen to make the a spacing of quartz 4.902 88 Å, derived from the measurements of Bergqvist (1930). However, evidence has been accumulating which suggests that this is small by about one part in twenty thousand, so it appears preferable to use the angle obtained from measurement. There seems to be no difficulty in measuring the parts of the camera with accuracy sufficient to obtain the angle within about 0.005° , which corresponds to a systematic error in the lattice parameter of less than one part in fifty thousand.

The lattice parameter was calculated for each reflection above about 50° , plotted against $\sin^2 \theta$ and extrapolated to $\sin^2 \theta = 1$. To allow for refraction the

The spacing and expansion coefficient of aluminium

Temp.	Films.	d_{obs}	d_{calc}	Diff.	$10^6 \partial d / \partial t$	$10^6 \partial d / \partial t$	$10^6 \alpha$	$10^6 \alpha$
0° C.	4	4.0390 ₆	4.0391 ₅	+0 ₉	90.1	89	22.3	22.0
25°	5	4.0413 ₄	4.0414 ₃	+0 ₉				
100°	1	4.0485 ₅	4.0485 ₉	+0 ₄	98.7	102.7	24.4	25.4
150°	1	4.0538 ₂	4.0536 ₃	-1 ₉				
200°	2	4.0591 ₅	4.0588 ₉	-2 ₆	107.3	107.5	26.4	26.5
300°	2	4.0700 ₅	4.0700 ₆	+0 ₁	116.0	113	28.5	27.8
400°	2	4.0819 ₈	4.0820 ₉	+1 ₁	124.6	122.2	30.5	29.9
500°	1	4.0946 ₈	4.0949 ₈	+3 ₀	133.3	133	32.6	32.5
600°	7	4.1086 ₇	4.1087 ₄	+0 ₇	141.9	146	34.5	35.5
650°	4	4.1161 ₈	4.1159 ₄	-2 ₄	146.2	153	35.5	37.2

The first values given for $\partial d / \partial t$ and α are from equation (2); the method of obtaining the others is discussed in the text.

spacings so found must be increased by an amount $(1-n)d$, where n is the refractive index of aluminium for the wave-length used. This correction is considered in more detail elsewhere (Edwards *et al.*, 1940; Wilson, 1940). For cobalt and copper radiations, those used in the present work, it amounts to 0.000 05 Å. and 0.000 03 Å. respectively.

About thirty exposures were made between 0° and 656°_3 C. They were ordinarily as close to 0° , 100° , 200° , . . . , 600° as could conveniently be arranged. The spacings were corrected to the nearest even 100° . The temperature difference was ordinarily about 5° , so only a rough value of the expansion coefficient was needed. The spacings so obtained are given in the table. The number of films

on which each is based is given in column 2 of the table. Nearly all are averages of more than one film. The mean deviation from the mean of repeated spacings is 0.000 1 Å. The spacing at 25°, 4.041 3₄ Å., is in good agreement with the 4.041 3₉ Å. of Jette and Foote (1935) and the 4.041 43 Å. of Ievinš and Straumanis (1936). The aluminium used was said by the makers to be 99.992 per cent pure.

§ 5. CALCULATION OF THE EXPANSION COEFFICIENT

To determine the coefficient of expansion, $\alpha = \frac{1}{d} \frac{\partial d}{\partial t}$, it is first necessary to obtain the derivative of d with respect to the temperature t . This is a matter of some difficulty. The values of d are nearly a parabolic function of t . By the method of least squares

$$d = 4.039 1_8 + 0.009 00_8 \left(\frac{t}{100} \right) + 0.000 431_7 \left(\frac{t}{100} \right)^2 \quad \dots\dots(1)$$

was found to be the best parabolic equation. It shows systematic deviations from the experimental points which are probably significant (columns 4 and 5 of the table). The slope of this equation is

$$10^6 \frac{\partial d}{\partial t} = 90.0_8 + 8.63_4 \left(\frac{t}{100} \right). \quad \dots\dots(2)$$

Values are given in column 6 of the table. The author is, however, of the opinion

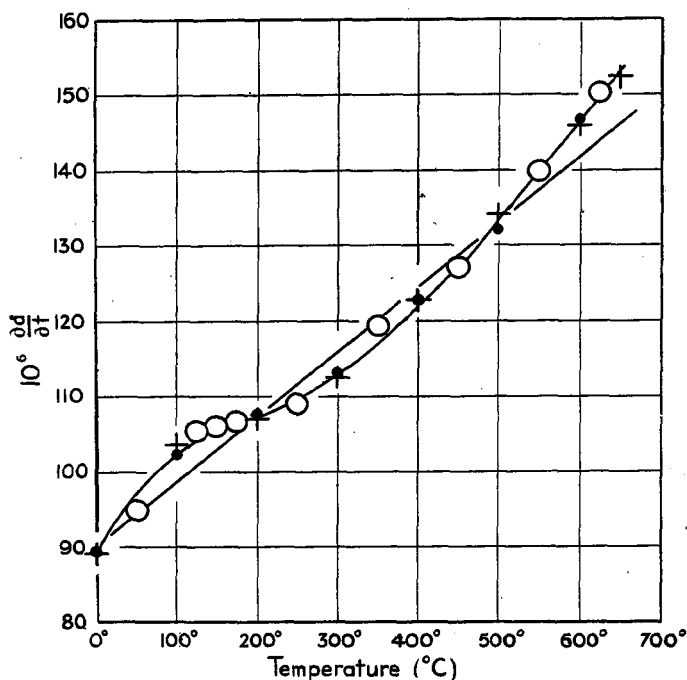


Figure 3. Slope of spacing-temperature curve.
Straight line, equation (2) ; for explanation of points see text.

that this equation does not give the best possible representation of $\partial d/\partial t$. It is plotted in figure 3 for comparison with values of $\partial d/\partial t$ obtained in other ways.

(1) The mean value of $\partial d/\partial t$ over intervals of 50° or 100° may be obtained by subtracting the values of d in the table and dividing by the temperature difference. These means are plotted as circles.

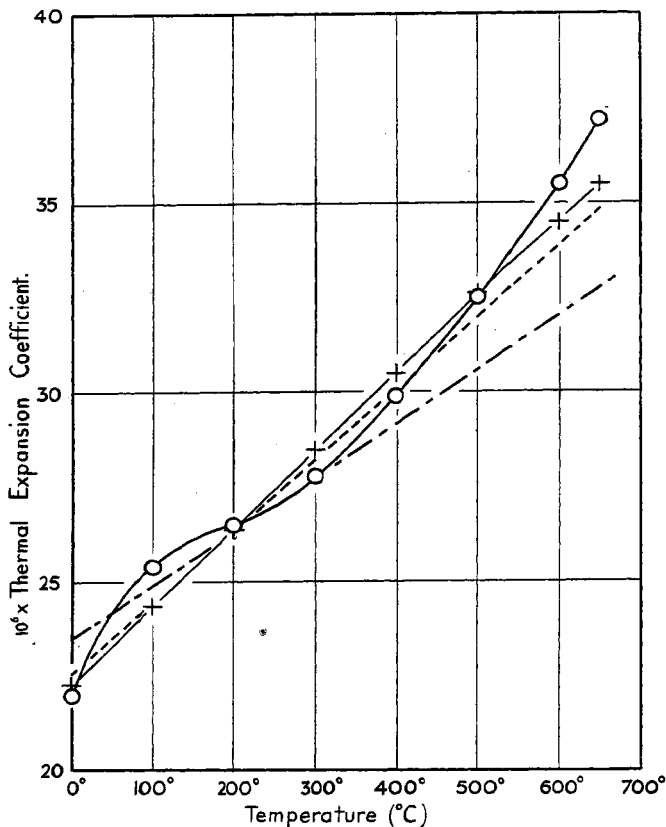


Figure 4. The thermal expansion coefficient of aluminium.

— — — — — Dittenberger ;
 - - - - - Hidnert ;
 x — x — x — x — x Equation (2) ;
 o — o — o — o — o Curve thought more probable by author.

(2) The mean value of $\partial d/\partial t$ between 0° and t° may be found,

$$\left(\frac{\partial d}{\partial t}\right) = \frac{d_t - d_0}{t}, \quad \dots\dots(3)$$

and the true value found from the relation

$$\frac{\partial d}{\partial t} = \left(\frac{\partial d}{\partial t}\right) + t \frac{\partial}{\partial t} \left(\frac{\partial d}{\partial t}\right). \quad \dots\dots(4)$$

The term $t \frac{\partial}{\partial t} (\overline{\partial d / \partial t})$ is fairly small compared with $\overline{\partial d / \partial t}$, so it introduces less uncertainty into the value of $\partial d / \partial t$ than there would be in the value of $\partial d / \partial t$ found directly. Values of $\partial d / \partial t$ from equation (4) are plotted as crosses.

(3) Successive groups of three values of d may be fitted by parabolas and the slope at the first, second and third temperatures plotted. These values are plotted as large dots.

The three methods agree in giving an S-shaped curve for $\partial d / \partial t$. This has been drawn in figure 4, and values read from it are given in column 7 of the table. It has not been thought worth while to reproduce the details of the calculations. Anyone interested should have little difficulty in repeating them.

Finally, the values of $\partial d / \partial t$ are divided by the corresponding values of d to give α , the true expansion coefficient. Column 8 gives α from equation (2), column (9) from the other methods. The latter is believed to be a better representation of the expansion coefficient of aluminium. In figure 4 the two sets of values are plotted for comparison with the values of α calculated from the equations of Dittenberger (1902) and Hidnert (1925). It will be seen that Hidnert's results agree closely with the values of α from equation (2). Unfortunately, he does not give the experimental results on which his equation is based, so it is not possible to see if they also would give an S-shaped curve on closer examination. Dittenberger's results are low above 300°, but he remarks that his equation is not a good fit for his experimental results. Esser *et al.* give only a small graph of their results, so it is difficult to compare them. Both d and α seem high, the latter by 8 per cent at 600° C.

§ 6. COMPARISON WITH GRÜNEISEN'S LAW

Grüneisen's Law, first deduced empirically (1908), states that α is proportional to C_p , the heat capacity at constant pressure. Various theories of the solid state lead to the relation $3\alpha = \rho\beta\gamma C_v$, where ρ is the density, β is the compressibility, γ is approximately a constant and C_v the heat capacity at constant volume. This relation, which does not differ essentially from Grüneisen's, may be derived by a method set out below which is strictly thermodynamical except for the assumption of the Debye or some similar expression for C_v . Its limits of validity should therefore be the same as those of the Debye expression for C_v .

The coefficient of cubic expansion is

$$3\alpha = \frac{1}{v} \left(\frac{\partial v}{\partial T} \right)_p = - \frac{1}{v} \left(\frac{\partial v}{\partial p} \right)_T \left(\frac{\partial p}{\partial T} \right)_v = \beta \left(\frac{\partial s}{\partial v} \right)_T. \quad \dots\dots(5)$$

The entropy s may be written as a function of v and T :—

$$s = \int_{v_0}^v \left(\frac{\partial s}{\partial v} \right)_T dv + \int_0^T \left(\frac{\partial s}{\partial T} \right)_v dT, \quad \dots\dots(6)$$

where the first integration is at $T=0$ and the second at constant volume v ; v_0 is

the volume at $p = T = 0$. Since $\alpha \rightarrow 0$ and β remains finite when $T \rightarrow 0$, the first integral vanishes (equation (5)). In the second integral $(\partial s / \partial T)_v = C_v / T$, so

$$s = \int_0^T \frac{C_v}{T} dT, \quad \dots\dots(7)$$

and, therefore,

$$3\alpha = \beta \left(\frac{\partial}{\partial v} \right)_T \int_0^T \frac{C_v}{T} dT. \quad \dots\dots(8)$$

The Einstein and Debye theories of heat capacity lead to the result that $C_v(v, T) = C_v(\theta/T)$, where θ is a function of v only. Substituting in (8) gives

$$\begin{aligned} 3\alpha &= \beta \int_0^T \frac{d\theta}{dv} \frac{C_v'(\theta/T)}{T^2} dT = -\frac{1}{v} \beta \frac{d \log \theta}{d \log v} \int_{\infty}^T C_v'(\theta/T) d(\theta/T) \\ &= \rho \beta \gamma C_v, \quad \dots\dots(9) \end{aligned}$$

on integrating and putting $\gamma = -d \log \theta / d \log v$, $\rho = 1/v$.

Neither the "law" $\alpha = \text{constant} \times C_p$ nor $3\alpha = \rho \beta \gamma C_v$ is well obeyed at high temperatures. Kelley (1934) gives the heat capacity of aluminium at 0°C . as 5.68 cal./gram atom $^\circ \text{C}$., at 600° as 7.61 cal./gram atom $^\circ \text{C}$. The ratio α/C_p at 0° is then 3.88, at 600° 4.62. It may be concluded, therefore, since the remainder of the argument is purely thermodynamical, that the assumptions on which the Debye theory is based are not satisfied by aluminium in the temperature range covered by this work.

§ 7. ACKNOWLEDGEMENTS

The author desires to express his thanks to Prof. Sir W. Lawrence Bragg, F.R.S., and Dr. A. J. Bradley, F.R.S., for their interest and advice, and particularly to Dr. H. Lipson for many helpful conversations on the design of the camera and the technique of spacing measurement. The camera was built by Mr. C. E. Chapman. The work was done while the author was an Overseas Science Research Scholar of the Royal Commission for the Exhibition of 1851.

REFERENCES

- BERGQVIST, O., 1930. *Z. Phys.* **66**, 494-498.
 BRADLEY, A. J. and JAY, A. H., 1932. *Proc. Phys. Soc.* **44**, 563-579; 1933. *Proc. Phys. Soc.* **45**, 507-522.
 DITTENBERGER, W., 1902. *Z. Ver. deutsch. Ing.* **46**, 1532-1536.
 EDWARDS, O. S., LIPSON, H. and WILSON, A. J. C., 1940. *Tables for Facilitating the Calculation of the Lattice Spacings of Cubic Crystals*. (The Crystallographic Laboratory, Cambridge.)
 ESSER, H., EILENDER, W. and BUNGARDT, 1938. *Arch. Eisenhüttenw.* **12**, 157-161.
 GRÜNEISEN, E., 1908. *Ann. Phys., Lpz.*, **26**, 211-216.
 HIDNERT, P., 1925. *Sci. Pap. Bur. Stand.* No. 497, 697-731.
 HUME-ROTHERY, W. and REYNOLDS, P. W., 1938. *Proc. Roy. Soc. A*, **167**, 25-34.
 IEVINŠ, A. and STRAUMANIS, M., 1936. *Z. phys. Chem. B*, **33**, 265-274.
 JAY, A. H., 1933. *Z. Kristallogr.* **86**, 106-111.
 JETTE, E. R. and FOOTE, F., 1935. *J. Chem. Phys.* **3**, 605-616.
 KELLEY, K. K., 1934. *Bull. Bur. Min. Wash.* No. 371.
 SEARS, J. E. and TURNER, A., 1941. *J. Sci. Instrum.* **18**, 17.
 WILSON, A. J. C., 1940. *Proc. Camb. Phil. Soc.* **34**, 485-489.